

Objective: Designing circuits which perform desired arithmetic operations

Problem 1-

problem specification- we need to design and implement a 1 bit full subtractor circuit using logic gates that takes two 4 bit numbers and a carry input bit as inputs and shows the difference of the numbers and a carry output bit as outputs.

Required Instruments-

<u>item</u>	<u>Quantity</u>
1. Trainer board	1 piece
2. IC extractor	1 piece
3. IC Hex inverter (7404)	1 piece
4. IC Quad 2 input AND (7408)	1 piece
5. IC Quad 2 input OR (7432)	1 piece
6. IC Quad 2 input Exclusive OR (7486)	1 piece
7. Wires	(a lot)
8. Hand book	1 piece

Truth Table:

Input			Output	
A	B	Bin	Bo	D
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Here $A = \text{input}$, $B = \text{input}$

$\text{Bin} = \text{carry input from previous step}$

$\text{Bo} = \text{carry output}$

$D = \text{Difference } [(A - B) - \text{Bin}]$

Kmap for D:

A \ B _{in}				
	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$D = AB'B_{in}' + A'B'B_{in} + ABB_{in} + A'B B_{in}'$$

$$= B'(AB_{in}' + A'B_{in}) + B(AB_{in} + A'B_{in}')$$

$$= B'(A \oplus B_{in}) + B(A \oplus B_{in})'$$

$$= A \oplus B_{in} \oplus B = A \oplus B \oplus B_{in}$$

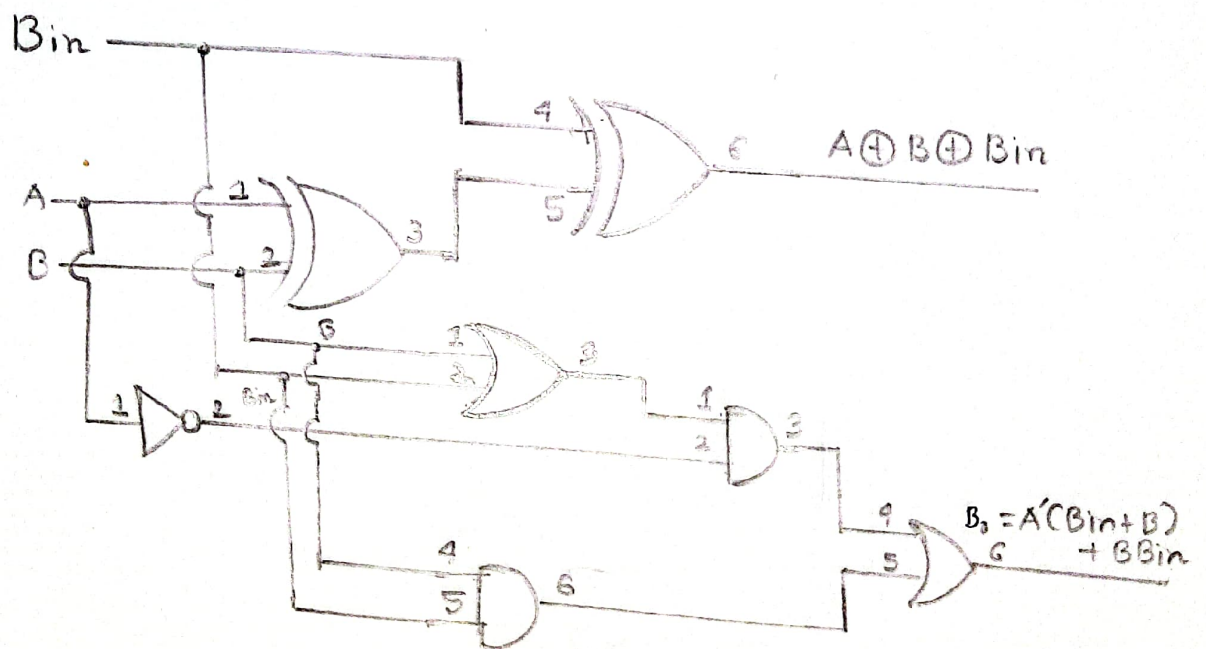
Kmap for B_0 :

$\begin{matrix} B_{in} \\ A \end{matrix}$	00	01	11	10
0		1	1	1
1			1	

$$B_0 = A'B_{in} + BB_{in} + A'B$$

$$= A'(B_{in} + B) + BB_{in}$$

Circuit Diagram:



Problem 2

Problem specification: We need to design and implement a 2 bit comparator using basic gates to compare two 2-bit numbers X and Y . The circuit provides 3 output lines to indicate, $x > y$, $x = y$ and $x < y$

Required instruments:

1. Trainer board 1 piece
2. IC extractor 1 piece
3. IC Hex inverter (7404) 1 piece
4. IC Quad 2 input AND (7408) 2 piece
5. IC Quad 2 input OR (7432) 2 piece
6. Wires A lot
7. Hand book 1 piece

Truth Table:

Input				Output		
X_1	X_0	Y_1	Y_0	$x < y$	$x = y$	$x > y$
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0 1	0	0
0	1	1	1	0 1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	1	0
1	1	1	1	0	0	1

Here,

 X_1 = Most significant bit of X X_0 = Least significant bit of X Y_1 = Most significant bit of Y Y_0 = Least significant bit of Y

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K-Map for $X < Y$:

$x_1 x_0 \backslash y_1 y_0$	00	01	11	10
00	0	1	1	1
01	0	0	1	1
11	0	0	0	0
10	0	0	1	0

$$\begin{aligned}
 X < Y &\Rightarrow x_1' y_1 + x_1' x_0' y_0 + x_0' y_1 y_0 \\
 &\Rightarrow x_1' y_1 + x_0' y_0 (x_1' + y_1)
 \end{aligned}$$

KMap for $X > Y$

$x_2 x_0 \backslash y_1 y_0$	00	01	11	10
00				
01	1			
11	1	1		1
10	1	1		

$$\begin{aligned}
 X > Y &\Rightarrow x_1 y_1' + x_0 y_1' y_0' + x_1 x_0 y_0' \\
 &= x_1 y_1' + x_0 y_0' (x_1 + y_1')
 \end{aligned}$$

K Map for $x=y$:

$x_1 x_0$	00	01	11	10
00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

$$\therefore x=y: x_1 x_0' y_1' y_0' + x_1' x_0 y_1' y_0 + x_1 x_0 y_1 y_0 + x_1 x_0' y_1 y_0'$$

$$= x_1' y_1' (x_0' y_0' + x_0 y_0) + x_1 y_1 (x_0 y_0 + x_0' y_0')$$

$$= (x_0' y_0' + x_0 y_0) (x_1 y_1 + x_1' y_1')$$

$$= \overline{(x_0 y_0' + x_0' y_0)} \cdot \overline{(x_1 y_1' + x_1' y_1)}$$

$$= \overline{x_0 y_0' + x_0' y_0 + x_1 y_1' + x_1' y_1}$$

Say $\Rightarrow x_1 y_1' + x_1' y_1 = Z \quad \therefore x_1 y_0' + x_0' y_0 = K$

$$\therefore = \overline{Z + K}$$

$$= \overline{Z + \overline{Z} K}$$

$$= \overline{x_1 y_1' + x_1' y_1 + (x_1 y_1 + x_1' y_1') (x_0 y_0' + x_0' y_0)}$$

$$= \overline{x_1 y_1' + x_1' y_1 + x_0 x_1 y_0' y_1' + x_0 x_1' y_0' y_1 + x_0' x_1 y_0 y_1' + x_0' x_1' y_0 y_1}$$

$$= \overline{x_0 y_1' (1 + x_0 y_0') + x_1 y_1 (1 + x_0' y_0) + x_0 x_1 y_0' y_1' + x_0 x_1' y_0' y_1 + x_0' x_1 y_0 y_1' + x_0' x_1' y_0 y_1}$$

$$= x_1 y_1' + x_0 x_1 y_0' y_1' + x_1' y_1 + x_0' x_1' y_0 y_1 + x_0 x_1 y_0' y_1 + x_0' x_1' y_0' y_1' + x_0' x_1 y_0 y_1 + x_0' x_1' y_0' y_1'$$

$$= x_1 y_1' + x_1' y_1 + x_0' x_1' y_0 y_1 + x_0' x_1' y_0' y_1' + x_0' x_1 y_0 y_1 + x_0' x_1' y_0' y_1' + x_0 x_1 y_0' y_1' + x_0 x_1' y_0' y_1' + x_0 x_1 y_0' y_1 + x_0 x_1' y_0' y_1'$$

$$= x_1 y_1' + x_1' y_1 + x_0' x_1' y_0 + x_0' y_0 y_1 + x_0 y_0' y_1' + x_0 x_1 y_0'$$

$$= x_1 y_1' + x_0' y_0 (x_1' + x_1) + x_1' y_1 + x_0 y_0' (x_1 + x_1')$$

$$= (x > y) + (x < y)$$

$$= \overline{(x > y)} \quad \overline{(x < y)}$$

Circuit Diagram:

